



Operate and maintain a 4-WD vehicle in safe working condition

D2.TTO.CL4.14

Assessor Manual



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Competency Based Assessment (CBA) – An Introduction for Assessors

Assessment is the process of identifying a participant's current knowledge, skills and attitudes sets against all elements of competency within a unit of competency.

Suggested Assessment Methods

For each unit of competency a number of assessment tools have been identified including:

- Work Projects
- Oral Questions
- Written Questions
- Third Party Statements
- Observation Checklists.

Instructions and Evidence Recording Sheets have been identified in this Assessment Manual for use by Assessors.

Alternative Assessment Methods

Whilst the above mentioned assessment methods are suggested assessment methods, the assessor may use an alternate method of assessment taking into account:

- a) The nature of the unit
- b) The strengths of participants
- c) The number of participants in the class
- d) Time required to complete assessments
- e) Time dedicated to assessment
- f) Equipment and resources required.

Alternate assessment methods include:

- Practical demonstrations
- Practical demonstrations in simulated work conditions
- Problem solving
- Portfolios of evidence
- Critical incident reports
- Journals
- Oral presentations
- Interviews
- Videos
- Visuals/slides/audiotapes

- Case studies
- Log books
- Projects and Role plays
- Group projects
- Recognition of Prior Learning.

Whilst there is no specific instruction or evidence collection documents for all the alternative assessment methods, assessors can record competency in the 'Other' section within the 'Competency Recording Sheet'.

Selection of Assessment Methods

Each assessor will determine the combination of Assessment Methods to be used to determine Competency for each Competency Unit on a student by student basis.

'Sufficient' evidence to support the 'Pass Competent'/'Not Yet Competent' decision must be captured.

In practice this means a minimum of 2 – 3 Assessment Methods for each candidate for each Competency Element is suggested.

At least one method should provide evidence of practical demonstration of competence.

The following assessment methods deemed to provide evidence of practical demonstration of competence include:

- Practical Work Projects
- Third Party Statement
- Observation Checklist.

Assessing Competency

Competency based assessment does not award grades, but simply identifies if the participant has the knowledge, skills and attitudes to undertake the required task to the specified standard.

Therefore, when assessing competency, an assessor has two possible results that can be awarded:

- 'Pass Competent' (PC)
- 'Not Yet Competent' (NYC).

Pass Competent (PC)

If the participant is able to successfully answer or demonstrate what is required, to the expected standards of the performance criteria, they will be deemed as 'Pass Competent' (PC).

The assessor will award a 'Pass Competent' (PC) if they feel the participant has the necessary knowledge, skills and attitudes in all assessment tasks for a unit.

Not Yet Competent' (NYC)

If the participant is unable to answer or demonstrate competency to the desired standard, they will be deemed to be 'Not Yet Competent' (NYC).

This does not mean the participant will need to complete all the assessment tasks again. The focus will be on the specific assessment tasks that were not performed to the expected standards.

The participant may be required to:

- a) Undertake further training or instruction
- b) Undertake the assessment task again until they are deemed to be 'Pass Competent'.

Regional Qualifications Framework and Skills Recognition System

The 'Regional Qualifications Framework and Skills Recognition System', also known as the 'RQFSRS' is the overriding educational framework for the ASEAN region.

The purpose of this framework is to provide:

- A standardised teaching and assessment framework
- Mutual recognition of participant achievement across the ASEAN region. This includes achievement in individual Units of Competency or qualifications as a whole.

The role of the 'RQFSRS' is to provide, ensure and maintain 'quality assurance' across all countries and educational providers across the ASEAN region

Recognition of Prior Learning (RPL)

Recognition of Prior Learning is the process that gives current industry professionals who do not have a formal qualification, the opportunity to benchmark their extensive skills and experience against the standards set out in each unit of competency/subject.

This process is a learning and assessment pathway which encompasses:

- Recognition of Current Competencies (RCC)
- Skills auditing
- Gap analysis and training
- Credit transfer.

Code of Practice for Assessors

This Code of Practice provides:

- Assessors with direction on the standard of practice expected of them
- Candidates with assurance of the standards of practice expected of assessors
- Employers with assurance of the standards maintained in the conduct of assessment.

The Code detailed below is based on the International Code of Ethics and Practice (The National Council for Measurement in Education [NCME]):

- The differing needs and requirements of the person being assessed, the local enterprise and/or industry are identified and handled with sensitivity
- Potential forms of conflict of interest in the assessment process and/or outcomes are identified and appropriate referrals are made, if necessary
- All forms of harassment are avoided throughout the planning, conducting, reviewing and reporting of the assessment outcomes

- The rights of the candidate are protected during and after the assessment
- Personal and interpersonal factors that are not relevant to the assessment of competency must not influence the assessment outcomes
- The candidate is made aware of rights and process of appeal
- Evidence that is gathered during the assessment is verified for validity, reliability, authenticity, sufficiency and currency
- Assessment decisions are based on available evidence that can be produced and verified by another assessor
- Assessments are conducted within the boundaries of the assessment system policies and procedures
- Formal agreement is obtained from both the candidate and the assessor that the assessment was carried out in accordance with agreed procedures
- The candidate is informed of all assessment reporting processes prior to the assessment
- The candidate is informed of all known potential consequences of decisions arising from an assessment, prior to the assessment
- Confidentiality is maintained regarding assessment results
- The assessment results are used consistently with the purposes explained to the candidate
- Opportunities are created for technical assistance in planning, conducting and reviewing assessment procedures and outcomes.

Instructions and Checklist for Assessors

Instructions

General instructions for the assessment:

- Assessment should be conducted at a scheduled time that has been notified to the candidate
- Facilitators must ensure participants are made aware of the need to complete assessments and attend assessment sessions
- If a participant is unable to attend a scheduled session, they must make arrangements with the Assessor to undertake the assessment at an alternative time
- At the end of the assessment the Assessor must give feedback and advise the participant on their PC/NYC status
- Complete the relevant documentation and submit to the appropriate department.

Preparation

- Gain familiarity with the Unit of Competency, Elements of Competency and the Performance Criteria expected
- Study details assessment documentation and requirements
- Brief candidate regarding all assessment criteria and requirements.

Briefing Checklist

- Begin the assessment by implementing the following checklist and then invite the candidate to proceed with assessment.

Checklist for Assessors

Prior to the assessment I have:	Tick (✓)	Remarks
Ensured the candidate is informed about the venue and schedule of assessment.		
Received current copies of the performance criteria to be assessed, assessment plan, evidence gathering plan, assessment checklist, appeal form and the company's standard operating procedures (SOP).		
Reviewed the performance criteria and evidence plan to ensure I clearly understood the instructions and the requirements of the assessment process.		
Identified and accommodated any special needs of the candidate.		
Checked the set-up and resources for the assessment.		
During the assessment I have:		
Introduced myself and confirmed identities of candidates.		
Put candidates at ease by being friendly and helpful.		
Explained to candidates the purpose, context and benefits of the assessment.		
Ensured candidates understood the assessment process and all attendant procedures.		
Provided candidates with an overview of performance criteria to be assessed.		
Explained the results reporting procedure.		
Encouraged candidates to seek clarifications if in doubt.		
Asked candidates for feedback on the assessment.		
Explained legal, safety and ethical issues, if applicable.		
After the assessment I have:		
Ensured candidate is given constructive feedback.		
Completed and signed the assessment record.		
Thanked candidate for participating in the assessment.		

Instructions for Recording Competency

Specifications for Recording Competency

The following specifications apply to the preparation of Evidence Gathering Plans:

- A Competency Recording Sheet must be prepared for each candidate to ensure and demonstrate all Performance Criteria and Competency Elements are appropriately assessed. This Sheet indicates how the Assessor will gather evidence during their assessment of each candidate
- This Competency Recording Sheet is located at the end of the Assessment Plan
- It is the overriding document to record competency
- The Assessor may vary the Competency Recording Sheet to accommodate practical and individual candidate and/or workplace needs
- Assessor must place a tick (✓) in the 'Assessment Method' columns to identify the methods of assessment to be used for each candidate
- Multiple Competency Elements/Performance Criteria may be assessed at the one time, where appropriate
- The assessor and participant should sign and date the Competency Recording Sheet, when all forms of evidence and assessment have been completed
- The assessor may provide and feedback or clarify questions which the participant may have in regards to the assessment grade or findings
- All documents used to capture evidence must be retained, and attached to the Competency Recording Sheet for each candidate for each Competency Unit.

Instructions for Different Assessment Methods

Specifications for Work Project Assessment

These guidelines concern the use of work projects.

The work projects identified in the Training Manuals involve a range of tasks, to be performed at the discretion of the Assessor.

Work project tasks can be completed through any form of assessment as identified in the Trainer and Trainee Manuals and stated at the start of this section.

Assessors should follow these guidelines:

- Review the Work Projects at the end of each 'Element of Competency' in the Trainee Manual to ensure you understand the content and what is expected
- Prepare sufficient resources for the completion of work activities including:
 - Time – whether in scheduled delivery hours or suggested time participants to spend outside of class hours
 - Resources – this may involve technical equipment, computer, internet access, stationery and other supplementary materials and documents
- Prepare assessment location (if done in class) making it conducive to assessment
- Explain Work Projects assessment to candidate, at the start of each Element of Competency. This ensures that participants are aware of what is expected and can collate information as delivery takes place

- Assessors can use the following phrase as a guide (where an 'X' is identified, please input appropriate information):
 "At the end of each Element of Competency there are Work Projects which must be completed. These projects require different tasks that must be completed.
 These work projects are part of the formal assessment for the unit of competency titled X:
 ■ You are required to complete these activities:
 a) *Using the 'X' method of assessment*
 b) *At 'X' location*
 c) *You will have 'X time period' for this assessment*
 ■ You are required to compile information in a format that you feel is appropriate to the assessment
 ■ Do you have any questions about this assessment?"
- Commence Work Project assessment:
 - The assessor may give time for participants to review the questions at this time to ensure they understand the nature of the questions. The assessor may need to clarify questions
 - Participants complete work projects in the most appropriate format
 - Participants must submit Work Project evidence to the assessor before the scheduled due date
- Assessor must assess the participant's evidence against the competency standards specified in each Element of Competency and their own understanding. The assessor can determine if the participant has provided evidence to a 'competent' standard
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Oral Question Assessment

These guidelines concern the use of oral questioning.

Assessors should follow these guidelines.

- Prepare Assessment Record for Oral Questioning. One record for each candidate:
 - Enter Student name
 - Enter Assessor name
 - Enter Location
- Familiarise self with Questions to be asked
- Prepare assessment location (table and chairs) making it conducive to assessment
- Explain Oral Questioning assessment to candidate, using the following phrase as a guide (where a 'X' is identified, please input appropriate information):
"These oral questions are part of the formal assessment for the unit of competency titled X.

There are X questions and you are required to answer all of them to the best of your ability and I will record whether or not you have answered correctly.

We have 60 minutes for this assessment.

- I will give you feedback at the end of the assessment
- Do you have any questions about this assessment?”
- Commence Oral Questioning assessment:
 - Complete Assessment Record for the Oral Questioning by:
 - a) Ticking PC or NYC, as appropriate
 - b) Entering ‘Remarks’ as required
 - c) Completing Oral Questioning within 60 minutes
- Complete Oral Questioning and provide feedback to candidate
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Written Question Assessment

These guidelines concern the use of written questioning.

Assessors should follow these guidelines:

- Familiarise self with Questions and Answers provided
- Print and distribute copies of ‘Written Questions’ for participants. Ideally this should take place with adequate time for participants to answer all questions before the expected due date
- Explain Written Questioning assessment to candidate, using the following phrase as a guide (where a ‘X’ is identified, please input appropriate information):

“These written questions are part of the formal assessment for the unit of competency titled X.

There are X questions and you are required to answer all of them to the best of your ability.

You may refer to your subject materials, however where possible try to utilise your existing knowledge when answering questions.

Where you are unsure of questions, please ask the Assessor for further instruction. This may be answering the question orally or asking the assessor to redefine the question.

We have X time for this assessment:

- The due date for completion of this assessment is X
- On this date you must forward the completed questions to the assessor by X time on the date of X
- Do you have any questions about this assessment?”
- The assessor may give time for participants to review the questions at this time to ensure they understand the nature of the questions. The assessor may need to clarify questions
- Participants may record written answers (where possible)
- Participants must submit the written answers to the assessor before the scheduled due date

- Assessor must assess the participant's written answers against the model answers provided as a guide, or their own understanding. The assessor can determine if the participant has answered the questions to a 'competent' standard
- Transcribe results/details to Competency Recording Sheet
- Forward/file assessment record.

Specifications for Observation Checklist

These specifications apply to the use of the Observation Checklist in determining competency for candidates.

Only an approved assessor is authorised to complete the Observation Checklist.

The assessor is required to observe the participant, ideally in a simulated environment or their practical workplace setting and record their performance (or otherwise) of the competencies listed on the Observation Checklist for the Competency Unit.

To complete the Observation Checklist the Assessor must:

- Insert name of candidate
- Insert assessor name
- Insert identify of location where observations are being undertaken
- Insert date/s of observations – may be single date or multiple dates
- Place a tick in either the 'Yes' or 'No' box for each listed Performance Criteria to indicate the candidate has demonstrated/not demonstrated that skill
- Provide written (and verbal) feedback to candidate – as/if appropriate
- Sign and date the form
- Present form to candidate for them to sign and date
- Transcribe results/details to Competency Recording Sheet for candidate
- Forward/file Observation Checklist.

This source of evidence combines with other forms of assessment to assist in determining the 'Pass Competent' or 'Not Yet Competent' decision for the participant.

Specifications for Third Party Statement

These specifications relate to the use of a relevant workplace person to assist in determining competency for candidates.

The Third Party Statement is to be supplied by the assessor to a person in the workplace who supervises and/or works closely with the participant.

This may be their Supervisor, the venue manager, the Department Manager or similar.

The Third Party Statement asks the Supervisor to record what they believe to be the competencies of the participant based on their workplace experience of the participant. This experience may be gained through observation of their workplace performance, feedback from others, inspection of candidate's work etc.

A meeting must take place between the Assessor and the Third Party to explain and demonstrate the use of the Third Party Statement.

To complete the Third Party Verification Statement the Assessor must:

- Insert candidate name
- Insert name and contact details of the Third Party
- Tick the box to indicate the relationship of the Third Party to the candidate
- Present the partially completed form to the Third Party for them to finalise
- Collect the completed form from the Third Party
- Transcribe results/details to Competency Recording Sheet for candidate
- Forward/file Third Party Statement.

The Third Party must:

- Record their belief regarding candidate ability/competency as either:
 - Pass Competent = Yes
 - Not Yet Competent = No
 - Unsure about whether candidate is competent or not = Not Sure
- Meet briefly with the assessor to discuss and/or clarify the form.

This source of evidence combines with other forms of assessment to assist in determining the 'Pass Competent' or 'Not Yet Competent' decision for the candidate.

A separate Third Party Statement is required for each Competency Unit undertaken by the candidate.

Competency standard

UNIT TITLE: OPERATE AND MAINTAIN A 4-WD VEHICLE IN SAFE WORKING CONDITION		NOMINAL HOURS: 60
UNIT NUMBER: D2.TTO.CL4.14		
UNIT DESCRIPTOR: This unit deals with skills and knowledge required to operate a <i>4-wheel drive</i> vehicle in a range of situations and provide for its basic maintenance while on tour		
ELEMENTS AND PERFORMANCE CRITERIA		UNIT VARIABLE AND ASSESSMENT GUIDE
Element 1: Operate a 4-wheel drive vehicle 1.1 Engage 4-wheel drive correctly and when appropriate 1.2 Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions 1.3 Provide safety and explanatory advice to passengers 1.4 Ensure minimal environmental impact when driving a 4-wheel drive vehicle 1.5 Comply with road safety and road use rules and regulations Element 2: Recover a 4-wheel drive vehicle 2.1 Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations 2.2 Optimise safety for passengers and protection of property		Unit Variables The Unit Variables provide advice to interpret the scope and context of this unit of competence, allowing for differences between enterprises and workplaces. It relates to the unit as a whole and facilitates holistic assessment. This unit applies to staff who operate a 4-wheel drive vehicle in a range of situations and provide basic maintenance while on tour, within the labour divisions of the hotel and travel industries and may include: 1. Tour Operation <i>Engage 4-wheel drive</i> refers to: • Selecting and engaging the 4-wheel drive according to manufacturer's instructions, including internal and external engagement • Selecting high or low range, or high or low model, depending on torque required • Selecting appropriate gear • Slowing or stopping to enable 4-wheel drive engagement, depending on vehicle type • Refraining from engaging optional 4-wheel drive on dry, hard, flat and firm surfaces • Protecting the safety of passengers and vehicle when engaging 4-wheel drive.

2.3 Inspect vehicle after recovery and <i>perform necessary maintenance</i> and repairs before proceeding Element 3: Provide basic maintenance for a 4-wheel drive vehicle 3.1 <i>Identify minor faults</i> 3.2 <i>Determine cause of faults</i> 3.3 <i>Conduct visual inspection of and operational tests on vehicle</i> 3.4 <i>Effect necessary repairs</i> 3.5 <i>Effect necessary preventative maintenance</i> Element 4: Report on 4-wheel drive operations 4.1 <i>Complete documentation</i> to record services and repair undertaken 4.2 <i>Notify management</i> of issues that require external attention	<i>4-wheel drive vehicle</i> may be: • Diesel or petrol powered • Made to manufacturer's instructions • Specially configured for touring conditions • Small capacity vehicles • Multi-seat units • Large or small coaches • Partial/part-time or permanent 4-wheel drive. <i>On-road conditions</i> refer to <i>sealed roads</i> and may include: • Slippery conditions, including rain and ice • Steep terrain • Snow. <i>Off-road conditions</i> refer to terrain that is not sealed and may include: • Steep inclines, including ascending and descending • Very soft ground • Rocky terrain • Sand • Water crossings • Mud. <i>Safety and explanatory advice</i> may be related to: • Advising the use of seat belts • Advising passengers to stay inside vehicle, including arms, hands and heads
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	• Explaining how 4-wheel drive transmission works • Stowing cargo appropriately, including re-packing cargo for off-road 4-wheel drive travel to attain a low centre of gravity • Taking pressure out of tyres. <i>Ensure minimal environmental impact should include:</i> • Adhering to designated tracks and routes in sensitive areas, parks or other prescribed areas • Assessing quality of tracks or intended routes before travelling on them • Minimising the effects of noise • Ensuring all required permits (including payment of fees), authorities and advice have been obtained and understood • Abiding by local prohibitions regarding entry by vehicles • Being vigilant to avoid damaging habitats, property, signs and other permanent and/or natural assets • Ensuring passengers do not litter the route and areas travelled • Ensuring passengers do not remove flora and/or fauna from the area • Not creating new tracks • Using passing bays, where provided • Leaving gates as they are found • Removing obstructions on tracks as opposed to driving round them • Not using trees as winch points if a recovery is required, or using tree protectors • Allowing animals to have right of way • Driving on beaches only on firm sand below high water marks.
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	<i>Techniques to recover 4-wheel drive vehicle</i> should relate to a variety of situations such as recovery from sand, mud, water and rocks, and may include: • Deflation of tyres before attempting to drive out • Selecting low range/ratio and low gear • Towing by another vehicle, including use of shackles, drag chains and snatch straps • Winching, including use of vehicle winch (electric, PTO, hydraulic and capstan), hand-operated winch, pulley blocks, winch extension straps and tree protectors • Using jacks, including kangaroo jack • Using under-vehicle inflatable bags • Using long handled shovel • Using portable tracks and/or sand ladders • Using front and rear recovery points. <i>Optimise safety for passengers and protection of property</i> may include: • Unloading items from the vehicle and removing to a safe and secure location • Disembarking passengers • Locating passengers at a safe distance (in a protected location) to avoid injury from flying parts that may occur during recovery. <i>Perform necessary maintenance</i> will most commonly apply to recoveries from water immersion and may include: • Checking no water in air intake • Remove spark plugs, or injectors and turn engine over to remove water • Replacing plugs, or injectors • Applying water displacement products to dry out high and low tension electrical circuits, as appropriate
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	• Removing visible debris • Re-aligning panel damage to enable free movement of wheels and steering • Making note to have differential and transmission checked and/or serviced at nearest opportunity. <i>Identify minor faults</i> may include: • Talking with co-drivers to identify type and nature of fault, including symptoms and indicators, frequency/conditions under which fault occurs/presents • Test driving vehicle to diagnose fault • Running engine of stationary vehicle to diagnose fault. <i>Determine cause of faults</i> may include: • Referring to service and operator manuals • Seeking advice from manufacturer, including over the phone assistance and radio communication • Eliminating factors from the scenario that is causing the fault on a sequential and structured basis while continuing ongoing testing • Referring to personal experience • Loading vehicle onto jacks to facilitate inspections • Use of applicable sensory appraisal, including sight, feel, smell and hearing • Monitoring on-board gauges and instruments. <i>Conduct visual inspection of and operational tests on vehicle</i> may include: • Checking for leaks • Checking condition of components, including hoses, belts, tyres, filters, radiator • Checking operation of indicators and lights
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	• Checking engagement point of brake pedal and clutch pedal • Checking parking brake operation • Checking for noise and vehicle emissions • Inspecting battery. <i>Necessary repairs</i> may include: • Replacing, adjusting or servicing items identified as unserviceable as a result of visual inspection and operational tests • Topping up and/or replacing fluids • Repairing punctures • Bleeding brakes • Bleeding a diesel engine • Replacing plugs. <i>Necessary preventative maintenance</i> will depend on the conditions in which the vehicle is being operated and may include: • Undertaking scheduled service tasks as recommended by the manufacturer • Undertaking specified additional service tasks as required by the host enterprise • Replacing component parts • Draining and replacing fluids, including topping up of fluids • Preparing the vehicle for conditions that are anticipated, including water and sand. <i>Documentation</i> may include: • On-board service records and logs. <i>Notify management</i> may include: • Verbal notification, including face-to-face, over the telephone, or via radio communication
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- Completing maintenance requisition
- Discussing the problem with specialists.

Assessment Guide

The following skills and knowledge must be assessed as part of this unit:

- The enterprise's policies and procedures in regard to the operation and maintenance of 4-wheel drive vehicles
- Principles of safe road use including on-road and off-road applications
- Knowledge of host country legislated and other requirements in relation to the service and maintenance of commercial passenger vehicles
- Principles of diagnosing vehicle faults
- Ability to use basic repair and maintenance tools, equipment and techniques
- Ability to apply knowledge of standard and basic maintenance and repair procedures
- Knowledge of vehicle manufacturer's instructions, specifications and recommended procedures for maintenance, trouble-shooting and repairs
- Knowledge of local road conditions
- Evidence of holding current and appropriate driver's license for the vehicle that is being driven.

Linkages To Other Units

- Establish and maintain safe touring conditions
- Drive large tour buses or coaches
- Manage operational risk
- Maintain tourism vehicles in safe and clean operational condition
- Drive various types of service vehicles
- Operate tours in remote areas.

	Critical Aspects of Assessment Evidence of the following is essential: • Understanding of host enterprise policies and procedures in regard to the operation and maintenance of 4-wheel drive vehicles • Understanding of the road rules and regulations of the host country and/or the road rules and regulations in which the 4-wheel drive vehicles are being driven • Proof of holding a current and valid driver's license to suit the 4-wheel drive vehicle • Demonstrated ability to perform required preventative maintenance and operational checks, as indicated on a supplied checklist, on a designated 4-wheel drive vehicle and provide written or verbal report on the condition of that vehicle • Demonstrated ability to diagnose at least three simulated, minor and unidentified faults on a nominated 4-wheel drive vehicle • Demonstrated ability to perform nominated and effective minor routine repairs to a designated 4-wheel drive vehicle in order to return the vehicle to a safe and effective operating condition • Demonstrated ability to drive a nominated 4-wheel drive vehicle in prescribed conditions so as to display at least seven of the following skill sets: ▪ Driving in sand ▪ Driving in mud ▪ Driving across rough terrain including rocky terrain ▪ Driving up and down steep inclines ▪ Driving through water ▪ Starting the vehicle, including key start and application of choke and hand controls to govern speed/engine revolutions ▪ Steering the vehicle and manoeuvring the vehicle, including uphill, downhill and on level ground
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	▪ Using all vehicle controls, instruments and indicators, including foot and hand-operated controls ▪ Positioning and parking the vehicle ▪ Stopping the vehicle, including uphill, downhill and on level ground. Context of Assessment Assessment must ensure: • Actual or simulated workplace application of vehicle maintenance and repair skills, and driving skills. Resource Implications Training and assessment must include the use of real vehicles, real tools and resources and a range of vehicles requiring a range of maintenance and repair provisions as well as the use of real vehicles in real or simulated driving conditions, situations and environments; and access to workplace standards, procedures, policies, guidelines, tools and equipment. Note: vehicle faults and problems may be simulated. Assessment Methods The following methods may be used to assess competency for this unit: • Observation of practical candidate performance • Portfolio of documents providing evidence of vehicle inspections, service, and repairs • Oral and written questions • Third party reports completed by a supervisor • Project and assignment work.
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	Key Competencies in this Unit <i>Level 1 = competence to undertake tasks effectively</i> <i>Level 2 = competence to manage tasks</i> <i>Level 3 = competence to use concepts for evaluating</i>		
	Key Competencies	Level	Examples
	Collecting, organising and analysing information	2	Check vehicles for faults and damage
	Communicating ideas and information	1	Convey safety information to passengers
	Planning and organising activities	1	Schedule vehicle serving; plan the best route to take given conditions that apply at the time
	Working with others and in teams	1	Drive safely and courteously with other road users; use passengers to assist in vehicle recovery
	Using mathematical ideas and techniques	1	Calculate quantities/volumes of fluids, air and other resources needed to service and repair vehicles
	Solving problems	2	Identify and diagnose faults
	Using technology	2	Drive vehicles and use on-board communication equipment; use servicing tools and equipment

Oral questions

Student name	
Assessor name	
Location/venue	
Unit of competency	Operate and maintain a 4-WD vehicle in safe working condition D2.TTO.CL4.14
Instructions	1. Ask student questions from the attached list to confirm knowledge, as necessary 2. Place tick in boxes to reflect student achievement (Pass Competent 'PC' or Not Yet Competent 'NYC') 3. Write short-form student answer in the space provided for each question.

Questions	Response	
	PC	NYC
1. Name a 4WD you have driven and explain how to engage four-wheel drive and describe situations when it is necessary to do so.	☐	☐
2. Name two off-road four-wheel drive situations and describe the driving techniques you use when driving in these conditions.	☐	☐

Questions	Response	
	PC	NYC
3. Tell me what you say when providing safety information to passengers who are in a 4WD you drive.	☐	☐
4. What do you do when driving a 4WD to minimise the impact of tour driving on the local environment?	☐	☐
5. How do you make sure you comply with local road safety and road use rules when driving a 4WD?	☐	☐

Questions	Response	
	PC	NYC
6. Tell me how you would recover a 4WD bogged in the mud, and one trapped in the sand.	☐	☐
7. What action do you take to optimise passenger safety when recovering a trapped/bogged 4WD?	☐	☐
8. What inspections and actions might you take after recovering a 4WD from being bogged in the mud to ensure there is no damage to the vehicle?	☐	☐

Questions	Response	
	PC	NYC
9. Name three minor faults that occur with a 4WD and tell me how you might identify these have occurred with the vehicle you are driving.	☐	☐
10. For the three faults identified in your answer to the previous question, what would you do to determine the cause of those faults?	☐	☐
11. Describe the standard visual inspection and operating tests you would perform on a 4WD before using it for a tour/at the start of a day.	☐	☐

Questions	Response	
	PC	NYC
12. For the faults being discussed, what action would you take to make the necessary repairs to each of them?	☐	☐
13. Identify a 4WD vehicle and describe the preventative maintenance required for this vehicle	☐	☐
14. What documentation might you complete to record the repairs and maintenance conducted on a 4WD you have driven on tour? What information would be included in these records?	☐	☐

Questions	Response	
	PC	NYC
15. Give me examples of three problems/issues you would report to management of instances where external attention is required for a 4WD.	☐	☐

Written questions

Operate and maintain a 4-WD vehicle in safe working condition – D2.TTO.CL4.14

Student Name: _____

Answer all the following questions and submit to your Trainer.

1. What is meant by a 'partial four-wheel drive'?

2. What is the difference between an AWD and an A4WD vehicle?

3. What happens when the diffs on a vehicle are locked and why may this be useful when driving a 4WD?

4. Identify five situations where 4WD may need to be selected.

5. Name a vehicle type and model and describe the process for selecting four-wheel drive.

6. What are presented in the notes as the Golden Keys to successful four-wheel driving?

7. Identify three requirements when driving a 4WD on dry, sealed roads.

8. Give six examples of preparations which may be necessary before taking a 4WD on tour in off-road conditions.

9. Describe the procedures for ascending a steep incline in a 4WD.

10. Describe the procedure for a safe stall recovery if a 4WD stalls when descending a steep incline.

11. When driving on hard sand should High range 4WD be selected, or Low range?

12. When driving through soft sand, should tyre pressure be increased or reduced? Why?

13. Describe the general procedures for driving a 4WD across rocky terrain.

14. Identify four pre-requisites which should be observed before making a water crossing in a 4WD.

15. Identify four situations where water crossings should be avoided.

16. Describe the general procedures for taking a 4WD across a water crossing.

17. Give six practical examples of responsibilities imposed on drivers of 4WD tour vehicles by Duty of Care obligations.

18. Give six examples of safety information which may be communicated to tour passengers by those who operate 4WDs.

19. Identify four possible ways tour 4WDs may impact the environment in which they travel.

20. Give eight techniques which have proved effective in assisting with environmental harm minimisation when driving a four-wheel drive vehicle.

21. List six generic compliance requirements for drivers of 4WD vehicles.

22. Describe four aspects of 'recovery gear knowledge' the driver of a tour 4WD should have.

23. Explain why it is important for 4WD drivers to use correctly rated recovery gear when recovering a trapped vehicle.

24. Give three possible uses of a long-handled shovel when recovering a 4WD from a trapped position.

25. When recovering a vehicle using a snatch strap, at what speed should the recovery vehicle move to initiate the recovery?

26. Answer 'True' or 'False' to the following statement: 'When recovering a vehicle using a snatch strap the trapped vehicle must be recovered on the first attempt, because the process cannot be repeated.'

27. Answer 'Yes' or 'No' to the following question: Should the driver of the trapped vehicle attempt to assist the recovery vehicle by attempting to drive forward when a snatch strap is used to retrieve a trapped vehicle?

28. When draping a cable dampener over a recovery where should it be located? Why?

29. Describe indicative considerations and actions when using an electric winch on a 4WD to undertake a self-recovery.

30. List three safety considerations when using a high-lift jack to raise a trapped 4WD.

31. Describe three actions to be undertaken *before* trying to drive/winch out a trapped vehicle when using sand ladders or light-weight tracks to assist with the recovery.

32. List four possible dangers which may be encountered when recovering a trapped vehicle.

33. What is the 'Danger Zone' in terms of recovering a trapped vehicle?

34. List four factors that determine the Danger Zone for the recovery of a vehicle.

35. What is the 'Golden Rule' when conducting a recovery or driving a four-wheel drive on-tour?

36. List four preparation activities which may be required when getting ready to perform a vehicle recovery.

37. Describe four important 'first responses' of the driver/tour crew when a 4WD becomes trapped.

38. Name six tasks which may need to be attended to after recovery of a 4WD on tour.

39. What are the four basic ways to identify vehicle faults?

40. Describe four SOPs for diagnosing/determining the cause of vehicle faults.

41. Describe four activities which could/should be included when conducting a visual inspection of a vehicle.

42. Describe indicative procedures for replacing a fuse on a vehicle.

43. Describe indicative procedures for replacing a coolant/radiator hose on a vehicle.

44. Identify one fluid which needs to be maintained on a vehicle and describe how to top it up.

45. Answer 'Yes' or 'No' to the following question: Can a deflated tyre be repaired using a plug kit if the sidewall of the tyre has been damaged?

46. Describe the general procedure for bleeding the brakes on a vehicle.

47. List six ways to determine the preventative maintenance requirements of a vehicle.

48. Identify eight examples of preventative maintenance tasks which may be required for an on-tour 4WD.

49. Give four reasons Tour Operators require service and repair documentation to be maintained by tour crew/tour vehicle drivers.

50. List four keys in the documentation process when completing on-tour service and repair documentation.

51. Identify four pieces of information required when completing a Vehicle Log/Vehicle Usage Log.

52. Describe three ways tour crews might notify management of tour vehicles issues requiring attention when they return at the end of a tour.

Answers to written questions

Operate and maintain a 4-WD vehicle in safe working condition – D2.TTO.CL4.14

The following are model answers only – Trainers/Assessors must use discretion when determining whether or not an answer provided by a Student is acceptable or not.

1. What is meant be a ‘partial four-wheel drive’?

Answer must address:

- Partial 4WD is also known as part-time 4WD.
- It means the vehicle is usually a 2-wheel drive vehicle when travelling normal roads/highways but has the capacity to become a 4WD when 4WD is activated or selected

2. What is the difference between an AWD and an A4WD vehicle?

- AWD = Full-time/permanent 4WD
- A4WD = Automatic 4WD where vehicle senses different individual wheel speeds and engages 4WD automatically it detects a need to do so.

3. What happens when the diffs on a vehicle are locked and why may this be useful when driving a 4WD?

- It drives both the axles (and hence both the wheels) with the same speed/power regardless of the road surface/condition.
- This facility is often a benefit in certain four-wheel driving conditions as it can give much better traction, drive, control and ability to get through sand, bogs and difficult terrain.

4. Identify five situations where 4WD may need to be selected.

- When tracks are muddy/boggy
- If the terrain is rough and rocky
- Where there is soft sand
- In the event of snow and ice
- Going up or down steep inclines
- When crossing creeks/rivers.

5. Name a vehicle type and model and describe the process for selecting four-wheel drive.

No model answer.

6. What are presented in the notes as the Golden Keys to successful four-wheel driving?

- Practice
- Experience.

7. Identify three requirements when driving a 4WD on dry, sealed roads.

- Use two-wheel drive/H2 only
- Leave extra distance between vehicles to allow for the extra braking distances required for heavier 4WD vehicles
- Take tight corners at a lower speed due to the relatively high centre of gravity for 4WDs
- Drive with headlights on at all times.

8. Give six examples of preparations which may be necessary before taking a 4WD on tour in off-road conditions.

- Loading the vehicle with necessary recovery gear
- Fitting protection and bash plates to the underside of the vehicle
- Fitting appropriate tyres
- Having tyre deflators and pumps
- Having warning/sand flags
- Having chains
- Fitting a snorkel
- Fitting a viscous fan
- Fitting extended differential breathers

9. Describe the procedures for ascending a steep incline in a 4WD.

- Inspect the path – looking for:
 - A suitable route through/across the track which presents itself
 - Surface condition
 - Approach angle
 - General angle/severity of the climb
 - Length of the incline
 - Possible recovery points that can be used
- Engage 4WD
- Engage diff locks
- Turn off air conditioner
- Begin the climb in a manner that reflects the findings of observation of the incline
- Stay square to the incline
- Work the wheel a little from side-to-side if the wheels start spinning
- Pause at the top of the incline – in order to:
 - Check the descent – if there is one
 - Check for oncoming vehicles.

10. Describe the procedure for a safe stall recovery if a 4WD stalls when descending a steep incline.

- Turn ignition off
- Apply foot brake
- Engage handbrake
- Depress clutch
- Select low gear
- Release clutch
- Slowly release handbrake
- Turn ignition on
- Release brakes
- Start vehicle and allow to continue down the slope
- Apply brakes emulating ABS as necessary to control descent
- Steer into skids.

11. When driving on hard sand should High range 4WD be selected, or Low range?

- High range.

12. When driving through soft sand, should tyre pressure be increased or reduced? Why?

- Reduced
- To increase grip/contact surface/footprint of the tyre with the sand – to spread the load over a bigger surface.

13. Describe the general procedures for driving a 4WD across rocky terrain.

- Inspect the path
- Reduce pressures in tyres
- Engage 4WD
- Engage diff locks
- Turn off air conditioner
- Begin in a manner that reflects the findings of observation – which may require, as appropriate:
 - Selecting Low range first or second gear
 - Driving slowly ('crawling') and smoothly to avoid wheel spin
 - Intelligently choosing the path to take to avoid traps, rocks and beaching of the vehicle
- Be prepared to ask passengers to get out of the vehicle – if patches of the route are dangerous
- Be prepared to ask passengers to act as spotters – to assist with providing directions on where to position wheels
- Try to stay on the high parts – straddling any ruts which may be present
- Re-inflate tyres when the rocky terrain has been negotiated.

14. Identify four pre-requisites which should be observed before making a water crossing in a 4WD.

- Know the vehicle and how deep water the water can be before the vehicle begins to float
- Know how fast the water is flowing – to determine safety and work out speed required and angle of attack
- Know the surface to be travelled – on the bottom of the river and on exit from the crossing
- Know where the air intake is – to make sure the vehicle does not ingest water
- Check-out the crossing – walk through the crossing to determine depth, identify obstacles and work out the path to be taken
- If in doubt, avoid the crossing.

15. Identify four situations where water crossings should be avoided.

- The current is too fast
- The water is too deep
- The approach or departure angles are too steep
- The approach or departure are too soft or otherwise too difficult
- The vehicle is not/cannot be adequately prepared.

16. Describe the general procedures for taking a 4WD across a water crossing.

- Let the engine cool down first – the steps below will assist with this
- Tie a tarpaulin across the front of the vehicle – to help create a bow wave
- Loosen the fan belt – to prevent damage to the radiator (unless vehicle is fitted with a viscous fan)
- Spray the motor with water repellent (WD-40) to help preserve the electrics
- Attach recovery straps to the vehicle – in case there is a need to recover the vehicle from the middle of the river: make sure these straps are secured above the water line
- Lower windows – as a safety factor
- Advise passengers not to wear seat belts
- Lower pressure in tyres – if exit is sandy/soft
- Select 4WD, low range – first or second gear depending on type and size of engine
- Lock the diffs
- Enter the water at a slow to moderate speed – so as to create a bow wave but not a 'splash' which will throw water up and over the bonnet/vehicle
- Drive slowly and evenly
- Avoid using the clutch – as water will enter the housing and cause damage, slippage
- Dry brakes after exiting the water.

17. Give six practical examples of responsibilities imposed on drivers of 4WD tour vehicles by Duty of Care obligations.

To ensure:

- The vehicle is safe/roadworthy when the tour departs
- The safety/roadworthiness of the vehicle is maintained through the trip
- All road rules and speed limits are obeyed
- The vehicle is driven in a safe and responsible manner
- Obvious risks are avoided or controlled
- Passengers are informed about potential risks/hazards
- Passengers comply with safety rules
- Cargo is stored or stowed appropriately.

18. Give six examples of safety information which may be communicated to tour passengers by those who operate 4WDs.

- To use/wear seat belts at all times – except when river/creek crossings are being undertaken
- To stay completely inside the vehicle at all times when the vehicle is moving
- To keep windows wound up when travelling through bush
- To wind windows down when undertaking river/creek crossings
- To use hand straps/holds when the vehicle is traversing uneven ground
- To follow the driver's instructions/directions at all times
- To refrain from altering any vehicle settings
- To secure loose items when the vehicle is in motion
- To refrain from throwing items/litter from the vehicle
- To refrain from behaviour that would be distracting to the driver of the vehicle
- To only get in and out of the vehicle when it has come to a full stop
- About on-board emergency items, first aid kits, recovery equipment, communication systems and emergency exits
- About Emergency Management Plans which have been developed – to guide action in the event of an emergency or accident.

19. Identify four possible ways tour 4WDs may impact the environment in which they travel.

- Disturbing local communities – and other visitors in the area
- Damaging/destroying flora – which can also lead to erosion of the land
- Causing noise pollution
- Contaminating fresh water sources
- Harming soil structure – by compacting, dislodging it and/or causing pot-holes

- Scaring, injuring or killing local wildlife.

20. Give eight techniques which have proved effective in assisting with environmental harm minimisation when driving a four-wheel drive vehicle.

- Adhering to designated tracks and routes in sensitive areas, parks or other prescribed areas
- Assessing quality of tracks or intended routes before travelling on them – and refraining from travelling on tracks which are:
 - Closed
 - Degraded or boggy
- Minimising the effects of noise – by:
 - Ensuring exhaust and muffler system is effective
 - Avoiding revving/racing engines
 - Staying clear of other users
- Ensuring all required permits (including payment of fees), authorities and advice have been obtained
- Abiding by local prohibitions – regarding, for example:
 - Entry by vehicles into nominated areas
 - Use of tracks under certain weather conditions
- Being vigilant when preparing, planning and driving – so as to avoid causing damage to:
 - Property and structures
 - Natural habitats of animals
 - Natural assets
- Ensuring proper behaviour of passengers
- Being patient
- Removing obstructions on tracks
- Not using trees as winch points if a recovery is required – or using tree protectors if no other suitable point is available
- Giving way to foot traffic on tracks – and allowing animals to have right of way
- Driving on beaches only on the firm sand below high water marks – so high tide will remove vehicle marks
- Adhering to weight limits for vehicles and trailers
- Taking care when driving through dry grass
- Not travelling at night.

21. List six generic compliance requirements for drivers of 4WD vehicles.

- Be properly licensed – according to:
 - The size of the vehicle
 - Type/class of vehicle
 - Whether it is manual or automatic
- Carry their licence – and present it on demand to authorities
- Ensure the vehicle is correctly registered – in accordance with:
 - The type/class of vehicle being used
 - The commercial nature of the vehicle
- Display proof of registration – to demonstrate current and relevant registration
- Ensure the vehicle complies with roadworthy requirements – as these apply to:
 - Vehicle safety
 - Passenger safety
- Obtain, display and comply with applicable special permits as/if required by local laws – for use of designated:
 - Parks
 - Tracks
 - Local roads
 - Beaches
 - Other routes/locations
- Comply with all applicable road rules – as these apply to:
 - Adhering to speed limits
 - Obeying traffic signs
 - Giving right of way to pedestrians and other vehicles – and passing and over-taking
 - Using indicators, lights and warning and safety devices
 - Demonstrating consideration and tolerance for other road users
- Comply with applicable fatigue management regulations – so the driver has sufficient:
 - Rest/sleep
 - Breaks from driving
- Comply with applicable environmental protection legislation – as this applies to:
 - Smoke emissions
 - Noise pollution

- Comply with relevant drink-driving legislation – and does not drive contrary to applicable blood alcohol limits as these apply to:
 - Age of the driver
 - Licence status
 - Type/class of vehicle being driven
- Comply with legislation prohibiting driving while under the influence of drugs.

22. Describe four aspects of ‘recovery gear knowledge’ the driver of a tour 4WD should have.

- Be aware of the gear available on/in the vehicle for recovery situations
- Know where the recovery gear is located
- Know the recovery points/hooks and mounting features/points of the vehicle – which are connected directly to the vehicle chassis to give optimum strength
- Prepare/attach any gear which is deemed necessary/possibly required for an upcoming situation
- Check the gear for safety and readiness for use – especially:
 - Checking for frayed cables and straps
 - Verifying winches which have been immersed in water on previous tours have been checked and serviced
 - Making sure items which may be required are readily accessible.

23. Explain why it is important for 4WD drivers to use correctly rated recovery gear when recovering a trapped vehicle.

- If incorrect/insufficiently rated gear (cables, straps, mounting points, winches, jacks) is used there is danger posed through:
 - The snapping/breaking of items
 - Items being ripped off vehicles
 - Jacks collapsing under weight
 - Winches failing under load.

24. Give three possible uses of a long-handled shovel when recovering a 4WD from a trapped position.

- Digging away sand or dirt – from under the wheels of vehicles
- Getting rid of debris from tracks/routes
- Removing unwanted soil or material from approach and departure angles of river crossings and steep inclines.

25. When recovering a vehicle using a snatch strap, at what speed should the recovery vehicle move to initiate the recovery?

At a medium/moderate rate

26. Answer 'True' or 'False' to the following statement: 'When recovering a vehicle using a snatch strap the trapped vehicle must be recovered on the first attempt, because the process cannot be repeated.'

False

27. Answer 'Yes' or 'No' to the following question: Should the driver of the trapped vehicle attempt to assist the recovery vehicle by attempting to drive forward when a snatch strap is used to retrieve a trapped vehicle?

Yes

28. When draping a cable dampener over a recovery where should it be located? Why?

In the middle third part of the strap

29. Describe indicative considerations and actions when using an electric winch on a 4WD to undertake a self-recovery.

- Always wear gloves
- Always follow manufacturer's instructions
- Unwind the cable off the winch – giving sufficient length to attach to the recovery point: add extra lengths of cable if necessary
- Place tree protector around the tree – and attach the winch cable with a rated shackle
- Drape a winch dampener over the cable – as a safety precaution
- Take up the slack on the cable between the tree and the trapped vehicle – guide the cable as it winds back on to the winch to ensure it rolls on evenly and smoothly
- Verify passengers are out of danger
- Start the recovery by engaging the winch – opinions vary and there are two basic options:
 - Allow the winch to do all the work and recover the vehicle on its own
 - Put the vehicle in 4WD Low range first or second gear and gently accelerate to assist the action of the winch
- Have a person (tour crew member) monitor the take up of the cable on the winch spool/drum – to ensure an even wind on
- Be aware of the heavy draw on electric power of the winch – so:
 - Leave engine running throughout the recovery operation
 - Minimise its use
 - Be prepared to use it in small bursts, stopping in between to run the engine and re-charge the battery
- Use a pulley block at the anchor point ('double line pull') to assist where necessary – which will halve the load on the winch: put a dampener over both cables

30. List three safety considerations when using a high-lift jack to raise a trapped 4WD.

- Provide solid surface
- Do not place head over handle
- Keep jack straight
- Connect to designated lift point
- Lift slowly.

31. Describe three actions to be undertaken *before* trying to drive/winch out a trapped vehicle when using sand ladders or light-weight tracks to assist with the recovery.

- Shovel away build up from in front of the tyre/s – if possible
- Raise the vehicle – if possible
- Position the ladders/tracks under and in front of the wheels
- Lower the vehicle onto the ladders/tracks
- Make sure vehicle is in 4WD Low range first gear.

32. List four possible dangers which may be encountered when recovering a trapped vehicle.

- Risk of the vehicle becoming flooded in a tidal river or beach situation as the water rises
- Danger of the vehicle slipping and rolling forward, backwards or sideways
- Chance of vehicle trapped in water being pushed over by the force of the water (current/flow) against its side
- Possibility of winch cable, snatch strap or tow rope breaking and springing back causing injury to people in vehicle or by-standers
- Passengers injuring themselves as they leave/get out of a stranded vehicle – and slip/slide on muddy, wet or otherwise slippery surfaces.

33. What is the ‘Danger Zone’ in terms of recovering a trapped vehicle?

The Danger Zone is the area in which passengers/people could be injured if something goes wrong with the recovery.

34. List four factors that determine the Danger Zone for the recovery of a vehicle.

- Type of recovery equipment used – which indicates possibility of (cable or rope or other) failure/s
- Direction of recovery to be attempted for the trapped vehicle
- Potential for vehicle to roll, slip, topple or fall – which may be related to:
 - Fall/lay of the land
 - Surface conditions
 - Physical position of the vehicle
- Effect of suction on cable/winch capacity
- Level of difficulty of the recovery.

35. What is the ‘Golden Rule’ when conducting a recovery or driving a four-wheel drive on-tour?

‘People safety is paramount’.

36. List four preparation activities which may be required when readying to perform a vehicle recovery.

- Check and prepare recovery gear which may be required – to ensure it is:
 - Quickly available
 - Fit for use should it be needed
- Check out/walk or otherwise inspect the route
- Re-locate items in the vehicle – in order to:
 - Alter the balance/weight distribution
 - Secure items
 - Give ready access to certain gear
- Advise passengers of identified threats
- Give necessary safety advice – such as:
 - Wind windows down
 - Keep doors unlocked
 - Keep seat belts on or off – depending on the situation.

37. Describe four important ‘first responses’ of the driver/tour crew when a 4WD becomes trapped.

- Take control of the situation
- Show leadership and confidence regarding the situation
- Avoid panic
- Inspect the situation to determine and gain first-hand knowledge and an understanding of the situation – rather than guess at what it might be
- Work out a plan of action
- Not rush – but take a deep breath and think things through
- Act – rather than react.

38. Name six tasks which may need to be attended to after recovery of a 4WD on tour.

- Checking there is no water in the air intake
- Inspecting tyres – for damage to tread and/or sidewalls
- Re-inflating tyres which have had pressure reduced
- Testing operation of systems and instruments
- Cleaning vehicle windscreen, mirrors and windows
- Replacing plugs, or injectors
- Applying water displacement products (such as WD-40) to dry out high and low tension electrical circuits
- Re-aligning panel damage to enable free movement of wheels and steering
- Making note to have differential and transmission checked (for water) and/or serviced at nearest opportunity
- Rolling cables and ropes back onto spools of winches
- Greasing and oiling nominated parts
- Removing visible debris – from areas and parts.

39. What are the four basic ways to identify vehicle faults?

- Talking with those who have operated/driven or been in a vehicle
- Test driving the vehicle
- Running stationary vehicle to induce fault
- Reading reports which have been prepared on the vehicle.

40. Describe four SOPs for diagnosing/determining the cause of vehicle faults.

- Analysing available information obtained from fault identification to provide list of tests to perform in order to:
 - Generate/cause the fault
 - Test options regarding what might be causing the problem
 - Eliminate possible causes in a structured and sequential, one-at-a-time basis
 - Identify causes
- Using diagnostic flowcharts
- Using on-board diagnostics (OBD)
- Referring to service and operator manuals
- Seeking additional advice from the manufacturer
- Referring to personal experience
- Loading vehicle onto jacks, stands or a hydraulic lift to:
 - Get a better look at the vehicle and be able to see 'up and under' it
 - Free up certain suspension components
 - Make it easier to manipulate other parts of the vehicle
- Using applicable sensory appraisal
- Monitoring on-board instrument panel
- Monitoring externally attached equipment.

41. Describe four activities which could/should be included when conducting a visual inspection of a vehicle.

- Looking at and under the vehicle and closely checking all aspects of the vehicle – while it is:
 - On the ground, and/or
 - On a hoist or over an inspection pit
- Lighting the areas being inspected
- Using a checklist – to:
 - Provide evidence the inspection occurred
 - Note defects and matters requiring attention
- Inspecting the vehicle in pairs – so that:
 - Issues can be analysed/discussed
 - A more effective inspection takes place – that is, there is less chance something will be missed if two people do the inspection
- Looking for signs of damage to the vehicle – as these signs can indicate:
 - Where a more detailed inspection needs to occur
 - Need for immediate action
 - Need to take later action
 - Possible cause or source of the damage
 - What may be needed/possible to prevent recurrence
- Checking for missing parts
- Noting leaking fluids
 - Identifying projections from the vehicle
 - Taking immediate action to rectify problems – which can be quickly and easily fixed on-the-spot, such as:
 - Clearing blockages
 - Tightening screws, nuts, bolts and fittings
 - Checking the condition/appearance of critical components of the vehicle (which may involve physical touching, moving, squeezing and removing of parts) such as:
 - Radiator hoses
 - Timing belts
 - Fan belts
 - Tyres including all spares
 - Oil, air and fuel filters
 - Battery/batteries, battery leads, posts, connections, brackets and mountings
- Checking fluid levels.

42. Describe indicative procedures for replacing a fuse on a vehicle.

- Locate fuse box/block
- Remove cover
- Read key on cover (or in manual) to determine the fuse to replace
- Remove fuse to be replaced
- Insert new fuse – press in position with finger
- Verify spare fuses are available for emergency use – where space provided to store them
- Test vehicle/accessory/light which was operating on this fuse.
- If fuse blows immediately then this usually indicates another problem (a short) and the vehicle may need to be taken to an auto-electrician.
- If light/accessory functions as required, replace fuse box/block cover.

43. Describe indicative procedures for replacing a coolant/radiator hose on a vehicle.

- Ensure engine has cooled before removing radiator cap
- Drain radiator – if necessary, using drain plug if fitted: there may be a requirement to retain the fluid to put back in the radiator.
- Remove bottom radiator hose from radiator if there is no drain plug – use
- Use pliers to release clamp holding hose in place – at front and back of top and bottom hose (if changing both hoses).
- It may be a good time to replace the thermostat while the top hose is off the vehicle.
- Use flat screwdriver to help loosen connection and twist hose to break seal and remove the hose
- Clean the parts to which the hose ends were secured – to remove visible debris
- Determine if new hose clamps need to be used to fit the new hose/s – use new clamps where indicated
- Position clamps (using pliers to open them up) near end of new hoses and replace hoses
- Re-attach clamps
- Fill radiator
- Run and check for leaks.
- Identify one fluid which needs to be maintained on a vehicle and describe how to top it up.

44. Answer 'Yes' or 'No' to the following question: Can a deflated tyre be repaired using a plug kit if the sidewall of the tyre has been damaged?

No.

45. Describe the general procedure for bleeding the brakes on a vehicle.

- Check the brake fluid reservoir – to make sure there is sufficient brake fluid available in the reservoir before starting the process.
- Make sure the level is monitored throughout the process so air is not re-introduced into the brake system.
- Have replacement/new brake fluid on hand – to top-up the reservoir as required
- Jack the vehicle up and remove the wheels – optional in some cases
- Attach brake bleeder bottle to brake nipple – open valve and depress brake pedal four to five times
- Close valve and tighten – top up reservoir with brake fluid
- Check for leaks by pressing brake pedal hard – re-tighten valve/nipple if there is a leak
- Remove brake bleeder bottle – discard fluid
- Repeat for remaining three wheels – monitoring level of brake fluid in brake fluid reservoir and topping-up as necessary.

46. List six ways to determine the preventative maintenance requirements of a vehicle.

- Take into account distance travelled and number of hours the vehicle has been used
- Conduct and note response to pre-start checks
- Comply with internal organisational SOPs
- Factor in the nature of roads/tracks travelled
- Use personal observation
- Refer to personal experience
- Research/read manufacturer requirements in relation to routine servicing and preventative maintenance.

47. Identify eight examples of preventative maintenance tasks which may be required for an on-tour 4WD.

- Cleaning the vehicle
- Undertaking scheduled service tasks as recommended by the manufacturer
- Undertaking specified additional service tasks as required by the host enterprise
- Replacing specific component parts – according to a nominated schedule
- Examining the tyres – and checking and/or adjusting the tyre pressures
- Topping up of fluids – as required based on inspection
- Tightening loose connections and fittings
- Replacing damaged or worn parts.

48. Give four reasons Tour Operators require service and repair documentation to be maintained by tour crew/tour vehicle drivers.

- Provide proof of inspections performed on vehicles while on tour
- Give a reference point for work which has to be performed on tour vehicles
- Allow comprehensive and accurate operational records to be maintained for each vehicle
- Enable accurate costing to be undertaken relating to the cost of operation of individual vehicles and types of vehicles
- Supply a database of information which can be provided to buyers when the tour vehicle is on-sold at the end of its useful life – to assist with proving regular maintenance and optimise re-sale value
- Help determine future repairs and maintenance which need to be provided to individual vehicles – based on the work they have done, and any service/maintenance performed on them
- Demonstrate tour crew/vehicle drivers' compliance with internal requirements – regarding daily inspections and scheduled/required maintenance and service
- Allow administration/management of the Tour Operator to reconcile receipts and invoices from external providers for vehicle expenses.

49. List four keys in the documentation process when completing on-tour service and repair documentation.

- Write clearly
- Be honest
- Cover all work done – be comprehensive
- Use only recognised abbreviations and acronyms
- Complete required documentation 'at the time' – as opposed to relying on memory to complete it at a later time.

50. Identify four pieces of information required when completing a Vehicle Log/Vehicle Usage Log.

- Name/title of tour
- Details of vehicle – usually by registration number, make and model
- Odometer reading/s – start and end of day
- Distance travelled – calculated from odometer readings
- Date/s
- Destinations and/or route taken – listing towns/locations: some Tour Operators may require odometer readings between nominated locations/destinations
- Name of driver/s – and signature
- Time/s – for travel (start and finish).

51. Describe three ways tour crews might notify management of tour vehicles issues requiring attention when they return at the end of a tour.

- Provide a comprehensive verbal report on relevant issues
 - Complete a vehicle maintenance requisition
 - Discussing the problem with specialists.
-

Observation checklist

Student name	
Assessor name	
Location/venue	
Unit of competency	Operate and maintain a 4-WD vehicle in safe working condition D2.TTO.CL4.14
Dates of observation	
Instructions	- Over a period of time observe the student completing each of the following tasks: - Operate a 4-wheel drive vehicle - Recover a 4-wheel drive vehicle - Provide basic maintenance for a 4-wheel drive vehicle - Report on 4-wheel drive operations - Enter the date on which the tasks were undertaken - Place a tick in the box to show they completed each aspect of the task to the standard expected in the enterprise - Complete the feedback sections of the form, if required.

Did the candidate	Yes	No
Element 1: Operate a 4-wheel drive vehicle		
Engage 4-wheel drive correctly and when appropriate	☐	☐
Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions	☐	☐
Provide safety and explanatory advice to passengers	☐	☐
Ensure minimal environmental impact when driving a 4-wheel drive vehicle	☐	☐
Comply with road safety and road use rules and regulations	☐	☐

Did the candidate	Yes	No
Element 2: Recover a 4-wheel drive vehicle		
Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations	☐	☐
Optimise safety for passengers and protection of property	☐	☐
Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding	☐	☐
Element 3: Provide basic maintenance for a 4-wheel drive vehicle		
Identify minor faults	☐	☐
Determine cause of faults	☐	☐
Conduct visual inspection of and operational tests on vehicle	☐	☐
Effect necessary repairs	☐	☐
Effect necessary preventative maintenance	☐	☐
Element 4: Report on 4-wheel drive operations		
Complete documentation to record services and repair undertaken	☐	☐
Notify management of issues that require external attention	☐	☐
Did the student's overall performance meet the standard?	☐	☐

Feedback to student and trainer/assessor**Strengths:****Improvements needed:****General comments:****Candidate signature****Date****Assessor signature****Date**

Third Party Statement

Student name:			
Name of third party:		Contact no:	
Relationship to student:	☐ Employer ☐ Supervisor ☐ Colleague ☐ Other Please specify: _____ Please do not complete the form if you are a relative, close friend or have a conflict of interest]		
Unit of competency:	Operate and maintain a 4-WD vehicle in safe working condition D2.TTO.CL4.14		
The student is being assessed against industry competency standards and we are seeking your support in the judgement of their competence. Please answer these questions as a record of their performance while working with you. Thank you for your time.			
Do you believe the trainee has demonstrated the following skills? (tick the correct response]	Yes	No	Not sure
Drives a 4WD under a range of tour and different on- and off-road conditions	☐	☐	☐
Preserves vehicle condition, passenger safety and the environment when driving or recovering the 4WD	☐	☐	☐
Uses different techniques to recover a 4WD from different bogged or trapped conditions	☐	☐	☐
Inspects 4WD after recovery and takes action to ensure it is safe to continue driving	☐	☐	☐
Identifies and determines cause of basic 4WD faults	☐	☐	☐
Undertakes basic repairs to 4WD	☐	☐	☐
Performs required preventative maintenance to 4WD	☐	☐	☐
Complete required documentation to record 4WD repairs and maintenance	☐	☐	☐
Notifies management of 4WD situations requiring attention by external providers	☐	☐	☐

Comments/feedback from Third Party to Trainer/Assessor:

Third party signature:

Date:

Send to:

Competency recording sheet

Name of Student		
Name of Assessor/s		
Unit of Competency	Operate and maintain a 4-WD vehicle in safe working condition	D2.TTO.CL4.14
Date assessment commenced		
Date assessment finalised		
Assessment decision	Pass Competent / Not Yet Competent (Circle one)	
Follow up action required (Insert additional work and assessment required to achieve competency)		
Comments/observations by assessor/s		

Place a tick (✓) in the column to reflect evidence obtained to determine Competency of the student for each Performance Criteria.

Element & Performance Criteria	Observation of skills	3rd Party Statement	Oral Questions	Written Questions	Work Projects	Other
Element 1: Operate a 4-wheel drive vehicle						
Engage 4-wheel drive correctly and when appropriate						
Drive 4-wheel drive vehicle in a variety of on-road conditions and off-road conditions						
Provide safety and explanatory advice to passengers						
Ensure minimal environmental impact when driving a 4-wheel drive vehicle						
Comply with road safety and road use rules and regulations						
Element 2: Recover a 4-wheel drive vehicle						
Use a variety of techniques to recover 4-wheel drive vehicle from bogged or trapped situations						
Optimise safety for passengers and protection of property						
Inspect vehicle after recovery and perform necessary maintenance and repairs before proceeding						

Place a tick (✓) in the column to reflect evidence obtained to determine Competency of the student for each Performance Criteria.

Element & Performance Criteria	Observation of skills	3rd Party Statement	Oral Questions	Written Questions	Work Projects	Other
Element 3: Provide basic maintenance for a 4-wheel drive vehicle						
Identify minor faults						
Determine cause of faults						
Conduct visual inspection of and operational tests on vehicle						
Effect necessary repairs						
Effect necessary preventative maintenance						
Element 4: Report on 4-wheel drive operations						
Complete documentation to record services and repair undertaken						
Notify management of issues that require external attention						
Candidate signature:			Date:			
Assessor signature:			Date:			

